Toward a four-dimensional Yang–Mills construction

Notebook 26: closing the support-weight compatibility test

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Abstract

An independent fixed grouping scale removes the unverified large-block decay premise in the inherited-interaction norm test. Physical averaging and its pivot chart keep their factor L. We carry quantum activities on current J-blocks and group Gaussian variables on current JL-blocks. After physical rescaling these return to next-current J-blocks. Known positive fixed-L spatial decay becomes any prescribed finite decay exponent in the grouped graph when J is large. The classical support expansions supply the finite cardinality reserves required for that regrouping. The tunable Gaussian density estimate then gives a fixed output support exponent, and both the new correction and inherited background term can return to the same positive carrier reserve. An explicit order of finite parameter choices satisfies the next-step support inequality. Analytic radii and norm constants remain explicit losses, so this closes support-weight compatibility without asserting a stable quantum induction.

Notebook 24 isolated a sufficient inequality that its original cell convention could not verify from positivity of fixed-L decay alone. Here the physical renormalization map, integration variables and sharp cutoff remain the same; the independent grouping changes how the support estimates are organized. The result applies to the homogeneous small-field step on a compatible cofinal family of periodic regulators.

1 Auxiliary grouping recovers a fixed quantum support weight

The physical averaging ratio and the size of a Gaussian expansion cell need not coincide. We keep the exact source ratio L fixed and group its Gaussian cells by a second integer J. This gives a finite-step recovery of one fixed connected-carrier exponent, without assigning arbitrary reserve to inherited quantum activities. It does not give contraction of activity amplitudes, stable analytic radii, or a quantum RG induction. All constants below are uniform in old refinement M and compatible periodic volume after the finite parameters L, J have been chosen.

1.1 The four graphs and parameter order

Fix the physical odd source ratio $L \geq 5$. On the current unit lattice, source blocks have side L and index graph Λ_{src}. This is also the next-current unit graph after rescaling lengths by L. Choose an odd auxiliary integer J, conveniently $J = L^j$. Gaussian integration cells group J^d source blocks, so their current physical side is JL; write their graph as Λ_G. After rescaling, these are J-blocks of the next current lattice.

At every stage, inherited quantum carriers are J-blocks of that stage's current unit lattice; call their graph Λ_Q. The two projections are

$$\rho_J : \Lambda_{\mathrm{src}} \longrightarrow \Lambda_G, \qquad \pi_L : \Lambda_Q \longrightarrow \Lambda_G. \tag{1}$$

Their fiber sizes are J^d and L^d, respectively. Thus the coarsening ratio of inherited carrier graphs remains L.

A centered odd JL cube is both a union of J^d centered L cubes and a union of L^d centered J cubes. The two smaller tilings need not be nested in one another; their common JL tiling is enough. Work on a cofinal family of compatible tori with periods divisible by JL. Taking $J = L^j$ preserves the original hierarchy on sufficiently large members. No claim for an incompatible short torus is needed.

The relevant dimensions are

$$D_G = J^d d(L^d - 1)\dim \mathfrak{g}, \quad D_{\mathrm{bg}} = d(JL)^d \dim \mathfrak{g}, \quad D_{\mathrm{in}} = dJ^d \dim \mathfrak{g}. \tag{2}$$

They count retained Gaussian coordinates, all background coordinates of a Gaussian macrocell, and background coordinates of an inherited carrier cell. Grouping is a coordinate rearrangement. It changes no source average, pivot, or cutoff.

Choose L first using the fixed weight test below. At that fixed L, the previous theorems supply positive covariance, classical precision and stationary-background decay exponents. Then choose J large enough for their finitely many conversion inequalities. Finally choose the product radius and coupling budgets. No large-L lower bound for the old decay exponents is required in this order.

1.2 Covariance and precision on macro Gaussian cells

Let the unchanged real baseline satisfy $\lambda I \le C_0 \le \Lambda_0 I$ and, on source blocks,

$$K_{\mathrm{src}}(\gamma_0) = \sup_i \sum_j e^{\gamma_0 d_{\mathrm{src}}(i,j)} \|(C_0)_{ij}\| < \infty, \qquad \gamma_0 > 0, \tag{3}$$

with the same column bound. These constants are uniform in M and volume for the fixed source ratio L. For i in macrocell A and j in macrocell B,

$$d_G(A,B) \le J^{-1} d_{\mathrm{src}}(i,j) + d. \tag{4}$$

It follows coordinatewise from block projection, including periodic distances. Since $\|(C_0)_{AB}\| \le \sum_{i\in A, j\in B} \|(C_0)_{ij}\|$,

$$K_G(\gamma) := \sup_A \sum_B e^{\gamma d_G(A,B)} \|(C_0)_{AB}\| \le J^d e^{\gamma d} K_{\mathrm{src}}(\gamma/J). \tag{5}$$

The same bound holds for columns. The spectrum is unchanged: the macro fibers are direct sums of the original Euclidean fibers, not averaged variables.

For any terminal set X, projection of a terminal spanning tree gives

$$\begin{aligned} \tau_G(\rho_J X) &\le 2J^{-1}\tau_{\mathrm{src}}(X) + d(|X| - 1), \\ \ell_G(\rho_J X) &\le 4J^{-1}\tau_{\mathrm{src}}(X) + 2d|X|. \end{aligned} \tag{6}$$

The first line uses (4) and $\ell_{\mathrm{src}} \le 2\tau_{\mathrm{src}}$; the second uses $\ell_G \le 2\tau_G$. These conservative constants suffice.

The classical precision has a strict symmetric expansion $A(V) = A_0 + \sum_X B_X(a|_X)$, with rows and columns in X. For some fixed $\sigma_0 > 0$ and any prescribed finite A_{card}, the earlier support theorem gives

$$\sup_i \sum_{X \ni i} \sum_{j\in X} e^{\sigma_0 \tau_{\mathrm{src}}(X) + A_{\mathrm{card}}|X|} \|(B_X)_{ij}\|_b \le C_A b. \tag{7}$$

The radius may shrink with the reserve. The exponent σ_0 can be fixed first because it comes from the fixed-L linear decay. The original conversion from current cells to source blocks is already included here.

Use the Gaussian parameters

$$\kappa_G = 2, \qquad m_G = 3, \qquad \gamma_G = \kappa_G + 2m_G = 8. \tag{8}$$

The strict Gaussian theorem requires $0 < \kappa_G < m_G$; in particular $m_G = 1$ would be invalid when $\kappa_G = 2$. Group precision labels by $X' = \rho_J X$ and their matrix rows and columns into macrocells. Equation (6) shows that its required macro precision norm at exponent γ_G is $O(b)$ if, in $d = 4$,

$$32/J \le \sigma_0, \qquad A_{\mathrm{card}} \ge 64. \tag{9}$$

The absolute block row sum costs at most $J^d C_A b$: sum the source row bound over the at most J^d source cells in one macrocell. Symmetry gives columns. Strict background support and individual symmetry survive grouping.

This reserve belongs to classical precision coefficients, where it is available. It is not a new assumption on quantum activities. The same argument groups the classical remainder and the local Haar, pivot-Jacobian and gauge multipliers at their required fixed target weights. Their original classical or local coefficients admit the necessary finite reserve. Their small fluctuation bounds remain $O_{L,J}(gR^3 + gR + g^2R^2)$ on the original retained window.

1.3 Regrouping the actual stationary support family

On source-block roots and next-current unit background labels, the matched one-step branch has strict families

$$a_i(w) = \sum_X u_{i;X}(w|_X), \qquad \sup_i \sum_X e^{\kappa_0 \tau_{\mathrm{src}}(\{i\}\cup X) + A_{\mathrm{st}}|X|} \|u_{i;X}\|_r \le C_{A_{\mathrm{st}}} r, \tag{10}$$

for some $\kappa_0 > 0$ and any prescribed finite A_{st}, with the corresponding smaller radius. The output at i is the whole current-background fiber of one source block.

Group output roots and background labels by ρ_J. For macro root I and macro support Z, its i-th source component is $\sum_{X:\rho_J X = Z} u_{i;X}$ for $i \in I$. Take the maximum over this whole macro output fiber before the support sum, and bound that maximum by the sum over $i \in I$. This costs J^d. The rooted projection inequality is

$$\tau_G(\{\rho_J i\} \cup \rho_J X) \le 2J^{-1} \tau_{\mathrm{src}}(\{i\} \cup X) + d|X|. \tag{11}$$

Consequently, if $8/J \le \kappa_0$ and $A_{\mathrm{st}} \ge 16$,

$$\sup_I \sum_Z e^{4\tau_G(\{I\}\cup Z)} \|u_{I;Z}\|_r \le J^d C_{A_{\mathrm{st}}} r =: C_{\mathrm{st}} r. \tag{12}$$

This is the actual stationary background, with macro spatial exponent $\kappa_{\mathrm{st}} = 4$. Its classical arbitrary reserve paid (11). No quantum reserve was spent. The later Taylor expansion of a general activity still counts the full dimension D_{bg}.

1.4 New fluctuation and normalization coefficients

The macro strict Gaussian theorem now has output reserve

$$a_G = \min\{1/2, \kappa_G/4\} = 1/2. \tag{13}$$

Choose $\theta = 1/(4D_G)$, so $\log \mathcal{K}_\theta \le 5/8$. The macro precision norm is $O(b)$, hence its strengthened θ-dependent smallness conditions can be imposed by shrinking b with L, J fixed. The edge constants may be large but remain finite.

The actual cutoff is the product of the retained link/color balls. A bad macrocell implies $\|\xi_I\|_2 > R$, so its joint tail is bounded by the Euclidean-block tail at the same threshold R. The opposite good-event

bound involving the square root of the number of links cannot improve this bad threshold. The chart uses the original link supremum R.

Two external normalization terms must be included. The local pivot-delta term

$$\log \frac{\mathcal{D}(V)}{\mathcal{D}(1)} = -\sum_{c} \operatorname{Tr} \log \pi_c(V)$$

has fixed source range. After grouping, it has anchored $O_{L,J}(b)$ norm with any fixed macro support weight, including $1/2$; a macrocell contains only $O(J^d)$ original local terms. This follows from its local formula, not from regrouping an old quantum coefficient family.

For the Gaussian determinant, rerun its chain expansion on the macro baseline and precision labels. Let K_γ and ν_γ be their weighted row norms at $\gamma = 8$, with $q = K_\gamma \nu_\gamma < 1$. Then

$$\mathcal{Q}(V) = -\tfrac{1}{2} \log \det(A(V) A_0^{-1}) = \tfrac{1}{2} \sum_{n \geq 1} \frac{(-1)^n}{n} \operatorname{Tr}[(C_0 B)^n]. \tag{14}$$

Expand each $B = \sum_Y B_Y$, join every entire label Y by a fixed minimizing tree, and join covariance endpoint pairs by fixed shortest paths. Their cyclic union is a connected declared carrier H. If T is the sum of these tree/path lengths and b_1 the initial matrix index, then

$$|H| \leq T + 1, \qquad z \in H \implies d_G(z, b_1) \leq T. \tag{15}$$

The graph is connected and has at most T distinct edges; these inequalities include anchors on added covariance paths. For $a_Q = 1/2$ and $\delta_Q = 1$,

$$\mathbf{1}_{\{z \in H\}} e^{a_Q |H|} \leq e^{1/2} e^{-d_G(z,b_1)} e^{(3/2)T} \leq e^{1/2} e^{-d_G(z,b_1)} e^{\gamma T}.$$

The matrix trace is bounded by D_G times the product of block norms. At fixed b_1, alternating the weighted covariance and precision-label row sums bounds a length-n chain by $D_G q^n$; dropping the final closing constraint only enlarges this positive majorant. The precision weight pays the internal trees since $\tau(Y) \leq \ell(Y)$. Summing b_1 gives $S_1 = \sum_{z \in \mathbb{Z}^d} e^{-|z|_1}$. Thus the connected determinant coefficients satisfy

$$\sup_z \sum_{H \ni z} e^{|H|/2} \|\mathcal{Q}_H\| \leq \frac{D_G e^{1/2} S_1}{2} \sum_{n \geq 1} \frac{q^n}{n} \leq \frac{D_G e^{1/2} S_1}{2} \frac{q}{1-q}. \tag{16}$$

Each coefficient reads only its precision-label backgrounds, a subset of H. The baseline has no hidden background. Absolute convergence gives the equality and holomorphy. This is a fresh macro determinant proof with reserve $1/2$, not an attempted amplification of an old low-weight family. Field-independent determinants and powers of g remain the separately tracked vacuum normalization.

Together, the new Gaussian and normalization coefficients have reserve $1/2$ and a norm ϵ_{new} bounded by $O_{L,J}(b)$ plus the stated fluctuation-log bound. This concerns their support weight, not an all-scale smallness or contraction assertion.

1.5 Inherited interactions still project by the ratio L

Assume inherited connected activities on Λ_Q satisfy

$$\sup_i \sum_{H \ni i} e^{a_0 |H|} \|E_H\|_b \leq \epsilon_{\text{old}}, \qquad a_0 = 1/16. \tag{17}$$

The source fluctuation map still has ratio L. On the independent product background $\|a\| < b_- < b$, its exact increment obeys

$$\|\Delta E_H\| \leq u \|E_H\|_b, \qquad u = \frac{C_{\text{ch}}(L)|g|R}{b - b_- - C_{\text{ch}}(L)|g|R}, \tag{18}$$

when the chart and positive denominator conditions hold. Directional Cauchy gives this estimate without a J-dependent coordinate count.

Nonpivot outputs are bond-local. A pivot output reads two source L-blocks, which either lie in one macrocell or in adjacent macrocells. As both the source and quantum tilings refine the common JL macrocell, its declared carrier obeys

$$\begin{aligned} &Y(H) \text{ is connected}, && |Y(H)| \le \chi|\pi_L H|, \quad \chi = d+1, \\ &\mathbf{1}_{\{B\in Y(H)\}} \le \sum_{i\in H} \mathbf{1}_{\{d_G(B,\pi_L i)\le 1\}}. \end{aligned} \tag{19}$$

The safe anchor factor is $N_{\mathrm{near}}L^d$, with $N_{\mathrm{near}} = 2d+1$; there is no extra J^d. Connected-carrier compression in the quantum graph gives

$$|\pi_L H| \le 2^d + (2^d/k_L)|H|, \qquad k_L = 1 + \lfloor (L-3)/4 \rfloor. \tag{20}$$

The target Gaussian multiplier exponent is at most

$$B_{\mathrm{req}} = 4 + \log \mathcal{K}_\theta + \kappa_G \le 53/8. \tag{21}$$

Consequently the inherited interaction weight is paid if

$$\frac{\chi 2^d}{k_L}\frac{53}{8} \le \frac{1}{16}. \tag{22}$$

In dimension four this asks for $k_L \ge 8480$. For example $L = 33921$ gives equality and $L = 33925$ a strict margin. These crude numbers demonstrate a finite choice, not a useful numerical coupling range.

After grouping equal carriers, let $Q_Y = \sum_{H:Y(H)=Y} \Delta E_H$ and $V_{\max} = uN_{\mathrm{near}}L^d\epsilon_{\mathrm{old}}$. The exact multiplier estimate yields

$$\epsilon_{\mathrm{int}} \le e^{V_{\max}} u e^{(53/8)\chi 2^d} N_{\mathrm{near}} L^d \epsilon_{\mathrm{old}}. \tag{23}$$

Indeed $\|Q_Y\| \le V_{\max}$ and $\|e^{Q_Y} - 1\| \le e^{\|Q_Y\|}\|Q_Y\|$; (19) and (20) then give the anchored sum. The base and inherited factors may share a support as distinct labels. Under the previously proved quarter budgets their connected-log difference is at most $4C_{\mathrm{out}}\epsilon_{\mathrm{int}}$. The old fluctuation-independent sum $\sum_H E_H(a)$ stays outside this integral.

1.6 Transport of both kinds of background term

Choose a positive common radius b_{com} no larger than b_- and every domain of the new Gaussian, determinant and delta coefficients. In particular $b_{\mathrm{com}} \le b$; restricting the old activities to this smaller ball preserves their input norm. The inherited outer radius b cannot silently be assigned to new coefficients constructed only on b_-.

Use (12) and the transport parameters

$$\beta = 1/8, \quad a_{\mathrm{out}} = \mu = 1/16, \quad \delta_{\mathrm{tr}} = 1, \qquad t = \frac{C_{\mathrm{st}} r}{b_{\mathrm{com}}} \le \frac{1}{16 D_{\mathrm{bg}}}. \tag{24}$$

Then $\beta + \delta_{\mathrm{tr}} < \kappa_{\mathrm{st}} = 4$ and $D_{\mathrm{bg}} \log(1/(1-t)) \le 1/8$. New Gaussian and normalization coefficients already have carriers in Λ_G and reserve $1/2$. Identity-projection transport returns reserve $1/16$ and norm at most

$$S_1 \epsilon_{\mathrm{new}}. \tag{25}$$

Their output carriers are J-blocks of the next current lattice, exactly the type assumed at the next stage.

The old background term has carriers in Λ_Q and must instead use π_L. Each of its J-current output fibers is a restriction of the containing JL macro fiber, so it inherits the rooted bound anchored at $\pi_L(i)$. Compressed transport uses $c_0 = 2^d, c_1 = 2^d/k_L$ and the coordinate count D_{in}. For $d = 4$,

$$c_1\beta + D_{\text{in}} \log \frac{1}{1-t} \le \frac{2}{k_L} + \frac{1}{8L^d} < \frac{1}{16}, \tag{26}$$

because $D_{\text{in}}/D_{\text{bg}} = L^{-d}$. It too returns reserve $1/16$, with norm at most

$$e^{\beta 2^d} L^d S_1 \epsilon_{\text{old}} = e^2 L^d S_1 \epsilon_{\text{old}}. \tag{27}$$

Its field-independent constants are tracked separately as vacuum terms. No reserve beyond $1/16$ was assumed for it. Treating it as if it already had the Gaussian output reserve $1/2$ would have been incorrect.

Thus the full update has the same connected-carrier exponent $a_0 = 1/16$ and a one-step bound

$$\epsilon_{\text{next}} \le e^2 L^d S_1 \epsilon_{\text{old}} + S_1 \epsilon_{\text{new}}, \tag{28}$$

on the common smaller field domain. The inherited fluctuation effect is already included in ϵ_{new} through (23). This is recovery of a function-space weight, not contraction of its norm.

1.7 A finite choice of J and the remaining induction boundary

After fixing L in (22), choose $J = L^j$ large enough that

$$8/J \le \gamma_0, \qquad 32/J \le \sigma_0, \qquad 8/J \le \kappa_0. \tag{29}$$

Every exponent on the right is positive for that fixed L, uniformly in M. Choose classical reserves at least 64 for precision and 16 for the stationary family, and any additional finite amount needed for classical interaction grouping. Their established constructions permit this at a smaller radius. Hence the choices have a finite solution without any large-L asymptotic estimate. The quantum family always starts on J-current cells; no extra quantum reserve is spent to amplify a macro spatial exponent.

The analytic radius is still a real obstruction. In particular,

$$r \le \frac{b_{\text{com}}}{16 D_{\text{bg}} C_{\text{st}}}, \qquad C_{\text{st}} = J^d C_{A_{\text{st}}}, \tag{30}$$

must be imposed along with all chart and precision domains. An incoming radius may itself have been produced by a previous step. This theorem does not turn (30) into a self-map, nor compare it with the required running-coupling domains. The smoother pre-gauge representative and its analytic-germ extension have separate integral and activity-norm interfaces.

Similarly, (28) has a coefficient larger than one, and the normalization terms are $O(b)$, not automatically $O(g)$. Uniform amplitude control requires vacuum, relevant and marginal extraction and a coupling-flow estimate. Large-field pieces, mixed histories and observable bounds are unchanged obligations.

The result is therefore specific: fixed L, J and the exact source coordinates preserve a fixed positive connected support exponent through inherited insertion, normalized Gaussian integration, external normalization and stationary transport. This resolves the earlier circular support-weight test while leaving the analytic-domain and amplitude induction explicitly open.

References

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